Ana Lorena Ortiz Loyola

Frankfurt am Main, Germany

+49 176 62145797 • a01750283@tec.mx • LinkedIn

Data science student (Tecnológico de Monterrey, expected 2027) with practical experience in data analysis, modeling, and visualization using Python and R, plus hands-on financial-services experience through a data engineering internship at Deutsche Bundesbank (AnaCredit) involving automated data-quality and audit-style verification of regulatory data. Combines statistical rigor from multiple predictive-modeling projects with a strong interest in machine learning, generative AI, and cloud technologies, including hands-on use of AI-assisted development tools (Claude Code) to accelerate a personal data-pipeline project.

Core Competencies

- **Data Analysis & Visualization**: Python, R, Pandas, NumPy, Matplotlib, Seaborn, Tableau
- **Financial Services & Audit-Adjacent Experience**: Data quality verification, migration testing, regulatory reporting (Deutsche Bundesbank AnaCredit)
- **Machine Learning**: Scikit-Learn, Keras, TensorFlow, Random Forest, SVM, K-Means; strong interest in generative AI
- **Programming**: Python, R, SQL, Java, C++
- **AI-Assisted Development**: Practical experience with AI coding tools (Claude Code), used to accelerate development of an end-to-end banking data pipeline project
- **Cloud & Tools**: Azure Cognitive Services, AWS, Git

Professional Experience

- **Deutsche Bundesbank (AnaCredit)** — **Frankfurt am Main, Germany**
  Data Engineer / Intern — *04/2026–07/2026*
  - Developed automated migration tests (row counts, referential integrity, aggregate consistency) using Pandas DataFrames to verify identical business logic after data migration – audit-style validation of regulatory reporting data
  - Built automated error detection identifying statistical deviations (duplicate IDs, value-range violations) in large-scale reported banking data, with automated Jira ticketing for detected issues

Projects

- **Personal Project (GitHub), built with Claude Code**
  Banking Data Pipeline – Data Cleaning & Audit Log *05/2026*
  - Built an end-to-end pipeline for simulated bank transactions (150 accounts): normalized SQLite database with foreign-key integrity checks
  - Staged data cleaning with a fully auditable log recording table, column, old/new value, action, and timestamp – designed to be audit-ready
  - Produced quality visualizations and a structured exception report
- **Tecnológico de Monterrey**
  Predictive Modeling – Healthcare Optimization *10/2024–12/2024*
  - Analyzed 25+ years of national health data; built and validated predictive models (Random Forest, Neural Networks) and K-Means clustering
- **Tecnológico de Monterrey**
  Air Quality Analysis Mexico City – SVM Classification *04/2023–06/2023*
  - Built SVM models (linear and RBF kernels) to classify critical ozone levels from citywide NOx/O3 data; visualized and communicated findings
- **Tecnológico de Monterrey**
  Lattice-Based Cryptography Research *10/2024–02/2025*
  - Implemented and analyzed the complexity of lattice-based cryptographic algorithms (LWE, NTRU) in Python; contributed to a scientific publication

Education

- **Tecnológico de Monterrey** **Mexico**
  BSc in Data Science *2022–2027*
  Grade: 1.3 (Academic Excellence Scholarship; DAAD KOSPIE Scholarship). Topics: data analysis & statistics, machine learning & AI, numerical optimization.
- **Universität Göttingen** **Germany**
  Exchange Semester *2025–2026*

Languages

- German (C1), English (C1, TOEFL iBT 105/120), Spanish (native).

Publications

- Contributed to a scientific publication on lattice-based cryptography (LWE, NTRU security parameter and runtime analysis), with Prof. A. F. De Abiega L'Eglisse, Tecnológico de Monterrey (citation pending publication).

Honors and Awards

- **Academic Excellence Scholarship** – Tecnológico de Monterrey.
- **DAAD KOSPIE Scholarship**.
- **National Physics Olympiad** – Mexico (2020).
- **Honorable Mention, Physics Olympiad** (2021).

References

- Available upon request.